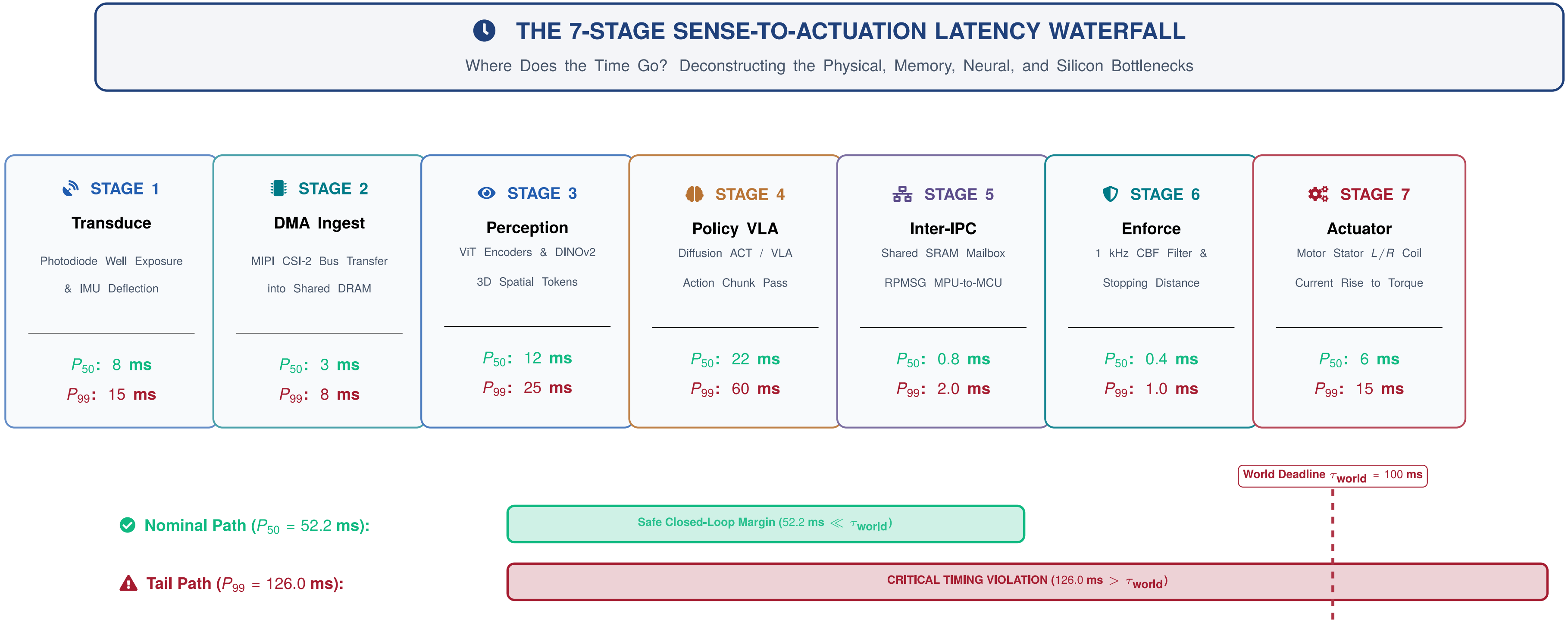
THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL
Where Does the Time Go? Deconstructing the Physical, Memory, Neural, and Silicon Bottlenecks
STAGE 1
Transduce
Photodiode Well Exposure & IMU Deflection
P_{50}: 8 ms
P_{99}: 15 ms
STAGE 2
DMA Ingest
MIPI CSI-2 Bus Transfer into Shared DRAM
P_{50}: 3 ms
P_{99}: 8 ms
STAGE 3
Perception
ViT Encoders & DINOv2 3D Spatial Tokens
P_{50}: 12 ms
P_{99}: 25 ms
STAGE 4
Policy VLA
Diffusion ACT / VLA Action Chunk Pass
P_{50}: 22 ms
P_{99}: 60 ms
STAGE 5
Inter-IPC
Shared SRAM Mailbox RPMSG MPU-to-MCU
P_{50}: 0.8 ms
P_{99}: 2.0 ms
STAGE 6
Enforce
1 kHz CBF Filter & Stopping Distance
P_{50}: 0.4 ms
P_{99}: 1.0 ms
STAGE 7
Actuator
Motor Stator L/R Coil Current Rise to Torque
P_{50}: 6 ms
P_{99}: 15 ms
World Deadline τ_{world} = 100 ms
Nominal Path (P_{50} = 52.2 ms):
Safe Closed-Loop Margin (52.2 ms $\ll \tau_{world}$)
Tail Path (P_{99} = 126.0 ms):
CRITICAL TIMING VIOLATION (126.0 ms $> \tau_{world}$)